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Filtering of WSS Processes

EEE 554 Lecture #19

Consider the situation in which a discrete-time Wide-Sense Stationary (WSS) process $x(k)$ is input to a linear time-invariant (LTI) system with impulse response $h(k)$.

The output process $y(k)$ has values:

$$y(k) = [x * h](k) = \sum_{n=-\infty}^{\infty} h(n)x(k-n)$$

i.e., the convolution of $x(\cdot)$ and $h(\cdot)$.

Pictorially: $x(k) \longrightarrow \boxed{\text{LTI: } h(k)} \longrightarrow y(k) = [x * h](k)$

Mean

In general,

$$\begin{aligned} \mu_y(k) &= E[y(k)] = E \left[\sum_n h(n)x(k-n) \right] \\ &= \sum_n h(n)E[x(k-n)] \\ &= \sum_n h(n)\mu_x(k-n) = [\mu_x * h](k) \end{aligned}$$

When $x(\cdot)$ is WSS, $\mu_x(k) = \mu_x$ is constant, so:

$$\mu_y(k) = \mu_x \sum_n h(n)$$

and μ_y is also constant.

Autocovariance

The autocovariance function of $y(\cdot)$ is given by:

$$\begin{aligned} C_y(k_1, k_2) &= \text{cov}(y(k_1), y(k_2)) \\ &= E[(y(k_1) - \mu_y(k_1))(y(k_2) - \mu_y(k_2))] \\ &= E \left[\sum_n h(n)(x(k_1 - n) - \mu_x) \sum_m h(m)(x(k_2 - m) - \mu_x) \right] \end{aligned}$$

$$= \sum_n \sum_m h(n)h(m)C_x(k_1 - n, k_2 - m)$$

If $x(\cdot)$ is WSS, $C_x(k_1 - n, k_2 - m) = C_x(K - n + m)$ with $K = k_2 - k_1$. So, in this case, $C_y(k_1, k_2)$ depends only on K ; thus $y(\cdot)$ is WSS with:

$$C_y(k) = \sum_n \sum_m h(n)h(m)C_x(k - n + m)$$

Spectral Density

With $x(\cdot)$ being WSS, $y(\cdot) = [x * h](\cdot)$ is also WSS with spectral density:

$$\begin{aligned} S_y(\omega) &= \sum_{k=-\infty}^{\infty} C_y(k)e^{-i\omega k} \\ &= \sum_k \sum_n \sum_m h(n)h(m)C_x(k - n + m)e^{-i\omega k} \end{aligned}$$

Let $l = k - n + m \Rightarrow k = l + n - m$:

$$\begin{aligned} &= \sum_n \sum_m h(n)h(m) \left(\sum_l C_x(l)e^{-i\omega(l+n-m)} \right) \\ &= \sum_n \sum_m h(n)h(m)e^{-i\omega n}e^{i\omega m} \sum_l C_x(l)e^{-i\omega l} \\ &= \left(\sum_n h(n)e^{-i\omega n} \right) \left(\sum_m h(m)e^{i\omega m} \right) S_x(\omega) \\ &= H(\omega)H^*(\omega)S_x(\omega) = |H(\omega)|^2 S_x(\omega) \end{aligned}$$

where $H(\omega) = \sum_{k=-\infty}^{\infty} h(k)e^{-i\omega k}$ is the frequency response of the LTI system.

Example: The RVs $x(k)$ are iid $\sim \mathcal{N}[0, \sigma^2]$ for $k \in \mathbb{Z}$. Then $E[x(k)] = 0$ for all k and:

$$C_x(k) = \begin{cases} \sigma^2 & k = 0 \\ 0 & k \neq 0 \end{cases} = \sigma^2 \delta(k)$$

Thus, for all ω :

$$S_x(\omega) = \sum_k \sigma^2 \delta(k)e^{-i\omega k} = \sigma^2$$

(i.e., $x(\cdot)$ is a white process).

If an LTI system has impulse response:

$$h(k) = \begin{cases} 1 & k = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

then:

$$H(\omega) = \sum_k h(k)e^{-i\omega k} = 1 + e^{-i\omega} = (1 + \cos \omega) - i \sin \omega$$

$$|H(\omega)|^2 = (1 + \cos \omega)^2 + \sin^2 \omega = 2 + 2 \cos \omega$$

If $x(\cdot)$ is passed through this system, the output process $y(\cdot)$ will have spectral density:

$$S_y(\omega) = |H(\omega)|^2 S_x(\omega) = \sigma^2(2 + 2\cos\omega) = 2\sigma^2(1 + \cos\omega)$$

Hence, $y(\cdot)$ is a lowpass process.

Moving Average (MA) Processes

Consider a causal FIR system. The corresponding system difference equation is:

$$y(k) = b_0x(k) + b_1x(k-1) + \dots + b_mx(k-m)$$

And the transfer function is thus:

$$H(z) = \frac{Y(z)}{X(z)} = b_0 + b_1z^{-1} + \dots + b_mz^{-m}$$

i.e., a polynomial of degree m in z^{-1} .

Definition: An m^{th} -order moving average (MA) process is the output of a causal FIR filter (as above) when the driving signal $x(k)$ is a zero-mean, finite-variance white process.

Notes: 1. If $x(\cdot)$ has variance σ^2 , then $S_y(\omega) = |b_0 + b_1e^{-i\omega} + \dots + b_me^{-im\omega}|^2\sigma^2$. 2. The filter weights $b_0, \dots, b_m$ are called the “MA coefficients” of $y(\cdot)$.

Autoregressive (AR) Processes

Consider a recursive linear system. The corresponding system difference equation is:

$$y(k) + a_1y(k-1) + \dots + a_ny(k-n) = x(k)$$

And the transfer function is thus:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{1 + a_1z^{-1} + \dots + a_nz^{-n}}$$

Recall that the filter is stable if all poles of $H(z)$ lie strictly within the unit disc in the z -plane.

Definition: An n^{th} -order autoregressive (AR) process is the output of a stable recursive filter (as above) when the driving signal $x(k)$ is a zero-mean, finite-variance white process.

Notes: 1. If $x(\cdot)$ has variance σ^2 ,

$$S_y(\omega) = |H(\omega)|^2 S_x(\omega) = \sigma^2 |H(e^{i\omega})|^2 = \frac{\sigma^2}{|1 + a_1e^{-i\omega} + \dots + a_ne^{-in\omega}|^2}$$

2. The filter weights $a_1, \dots, a_n$ are called the “AR coefficients” of $y(\cdot)$.